



Team Progress Update #2

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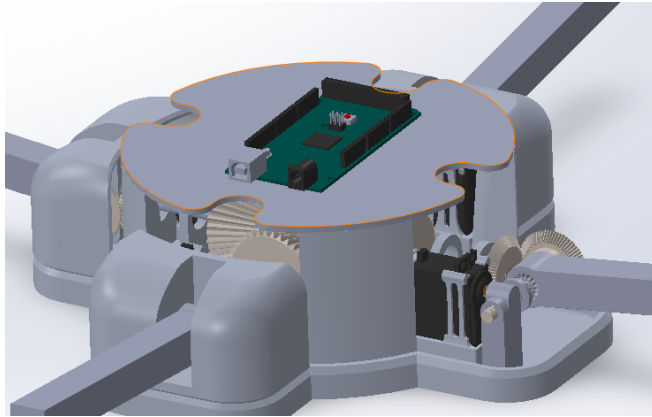
MIME-5504A

05/23/2025

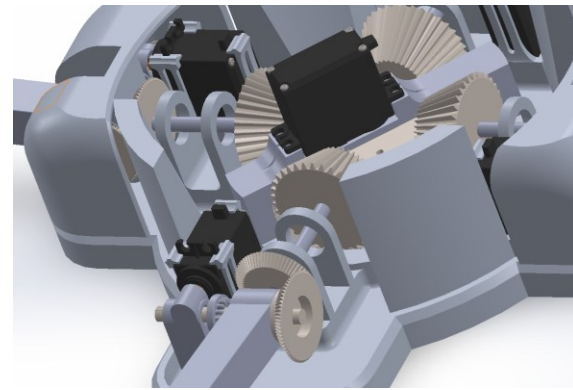
CAD Prototype 1: Black Box

CAD Progress:

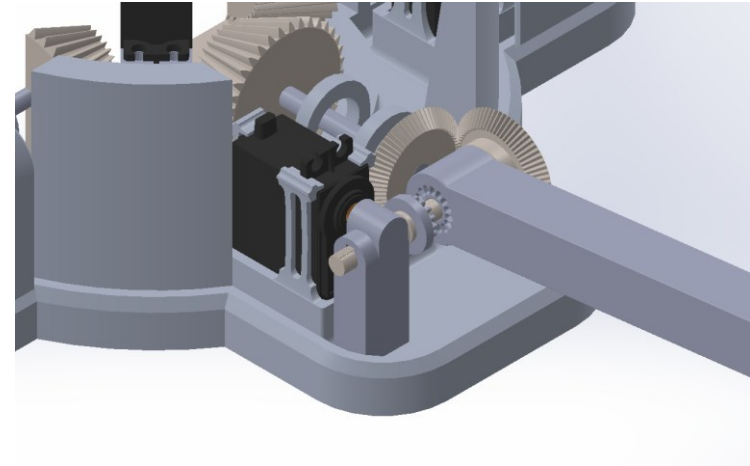
- Switchboard
- Gear Development
- McMaster-Carr BOM Inclusions



Black Box Module of Attachment

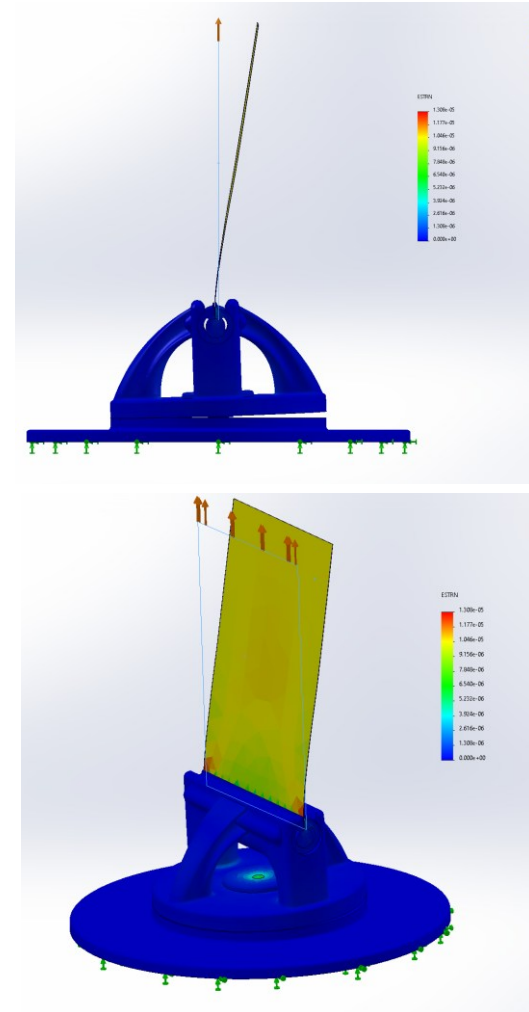
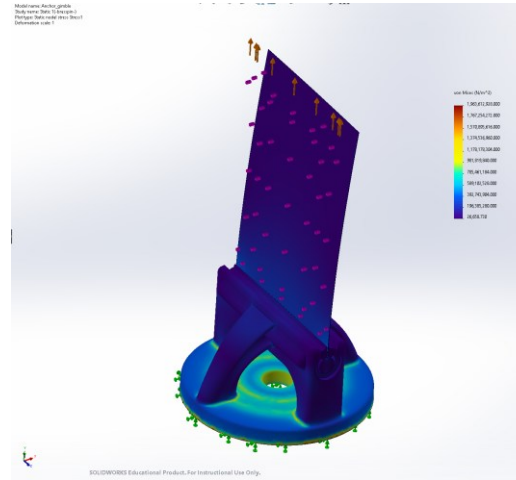
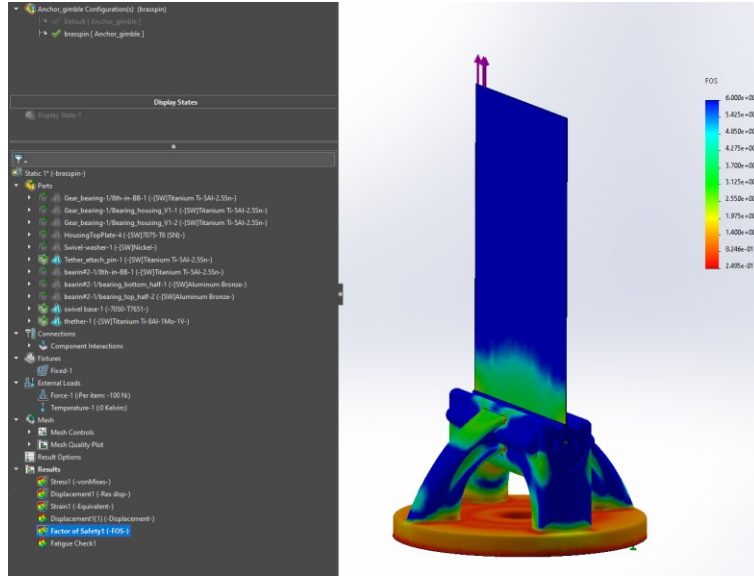


Gear Functionality Exploded View



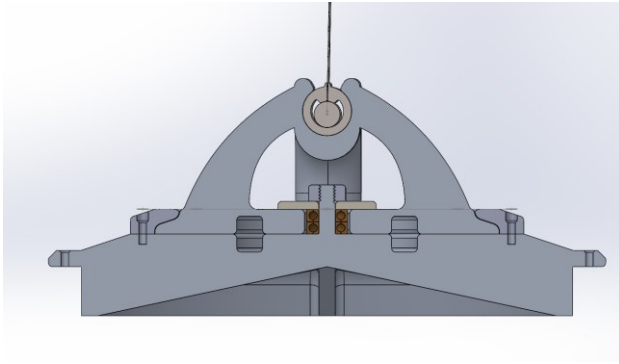
Cross-Sectional View of Black Box Module

FEA of Tether Gimbal Attachment Point - Highlights

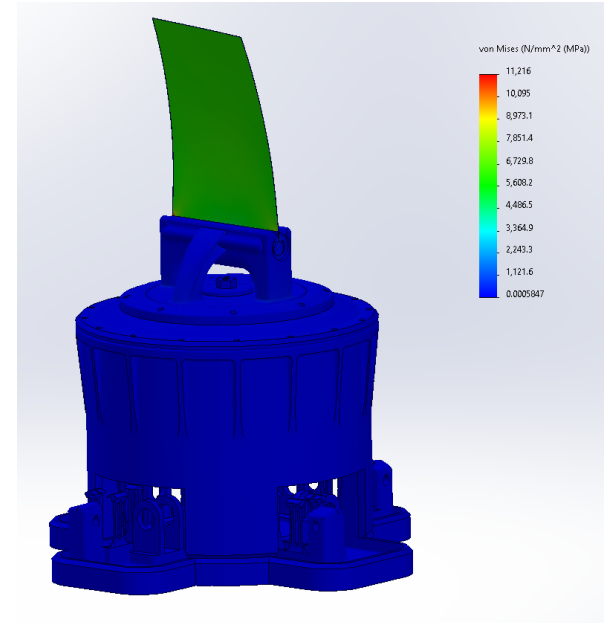
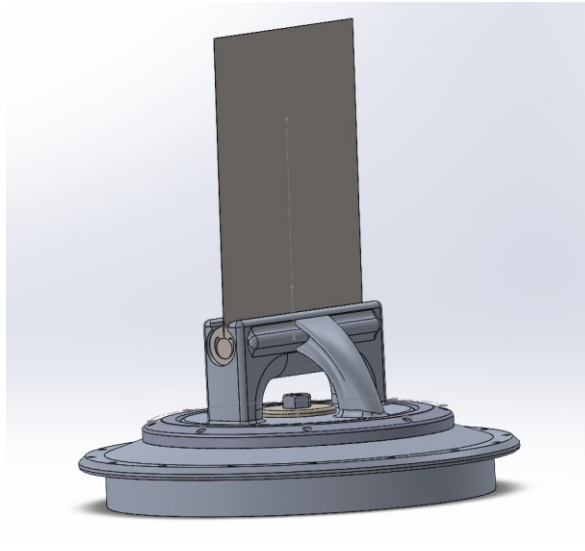


CR#	CR Description	Weight	Matching Engineering Specification
6	Strong enough material to perform mandatory functions & withstand varied impact conditions	10	Shear Strength ≥ 205 MPa, UTS ≥ 600 MPa, Yield Strength ≥ 385 MPa (Al 6061-T6)
9	Ability to effectively interface with tether device	5	Attachment point to relevant team concept

FEA Influenced Redesign



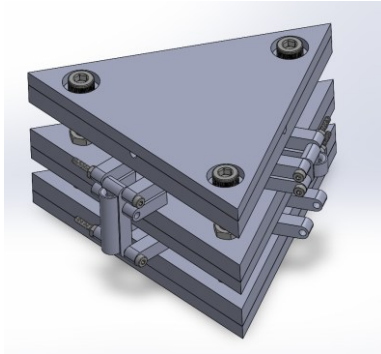
- Gimbal Constraining Ring
- Plate Support Ribbing
- Black Box Shell Support Ribbing



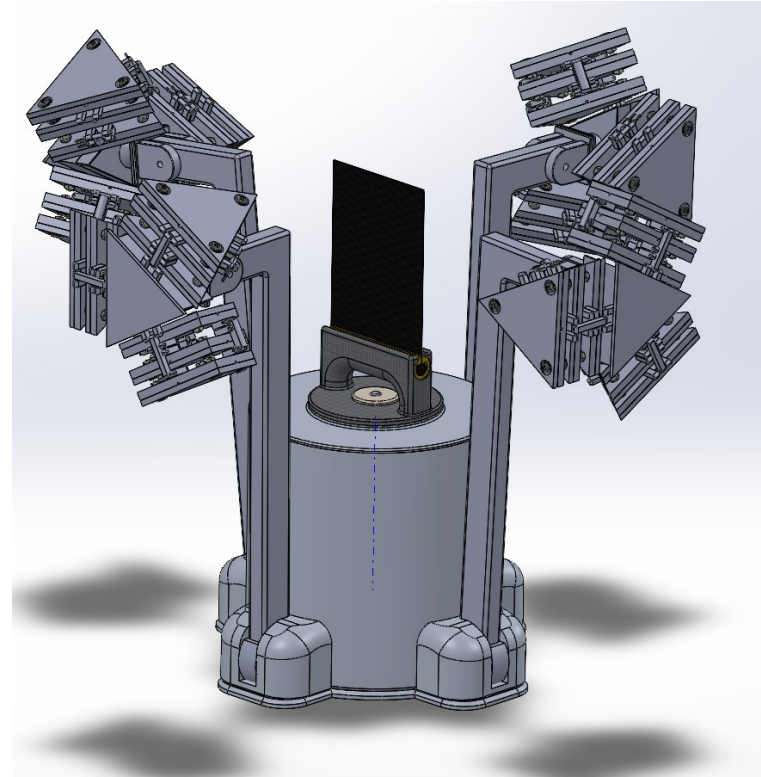
CAD Prototype 2: Feet

CAD Key Functions:

- Folding Mechanism
- Wire for expanding DoF
- McMaster-Carr BOM Inclusions



Unfolding Foot Module



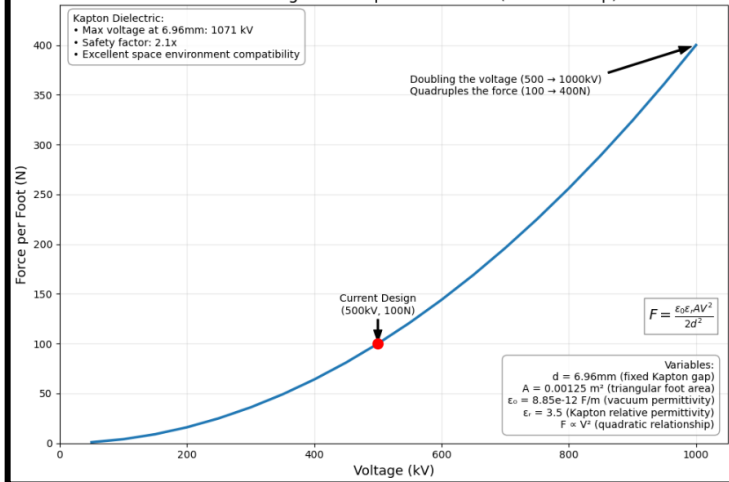
Overall Assembly With Feet and Legs

CR#	CR Description	Weight	Matching Engineering Specification
5	Able to self-anchor in near-zero gravity environments	10	Is able to land and anchor while experiencing <1% of Earth's gravity (-0.106 m/s ² minimum)
8	Ability to anchor with an uneven surface	15	Anchors can attach onto surfaces that have height differences of up to 10 meters, handle at least 34 degrees of tilt

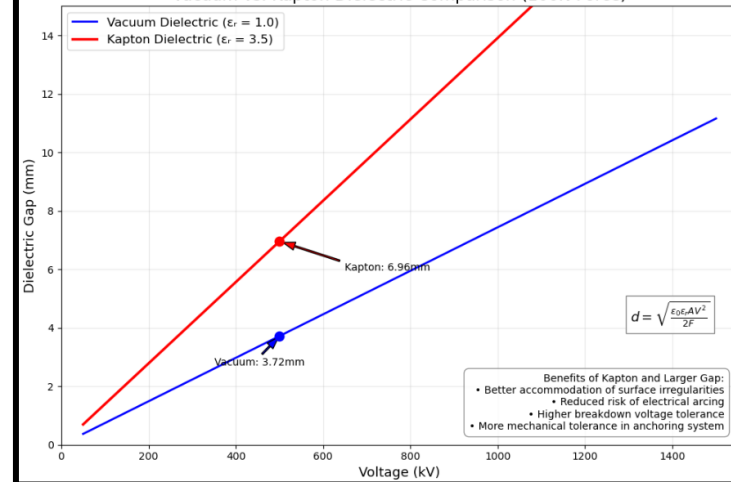
MBOM Progress & Purchases

Item Name	Brand	Quantity	Unit Cost	Total Cost	Price Checked With Receipt	Source	Date Purchased (Requested)	Date Received
12V Power Supply	ALITOVE	1	\$9.58	\$9.58	Yes	Amazon Link	4/4/2025	4/9/2025
51110 Thrust Ball Bearings 50mm x 70mm x 14mm Chrome Steel ABEC3 Single Row Roller 2pcs	uxcell	1	\$13.19	\$13.19	Yes	Amazon Link	4/4/2025	4/9/2025
20-Pack R4-2RS Bearings, ID 1/4 x OD 5/8 x 0.196 inch Small Ball Bearings	Donepart	1	\$15.99	\$15.99	Yes	Amazon Link	4/4/2025	4/9/2025
320pcs M3 Circuit Board Mounting Kit	Lystali	1	\$9.99	\$9.99	Yes	Amazon Link	4/4/2025	4/9/2025
2 Pack, 1" Ball Adapter with M10 x 1.25 x 15 Threaded Post Compatible with RAM Mounts B Size 1 Inch Ball Double Socket Arm	BROOVAN	2	\$17.14	\$34.28	Yes	Amazon Link	4/4/2025	4/9/2025
22 awg Silicone Electrical Wire (100ft)	Haerkin	1	\$16.98	\$16.98	Yes	Amazon Link	4/4/2025	4/9/2025
Pro 950 Polyimide Film Tape, 7500V Dielectric Strength, 36 yds Length x 1" Width	Pro Tapes	1	\$13.56	\$13.56	Yes	Amazon Link	4/4/2025	4/9/2025
[4-Pack] MG996R 55g Servo Motors	Deegoo FPV	1	\$17.99	\$17.99	Yes	Amazon Link	4/4/2025	4/9/2025
40KG Digital Servo Motor 360 Degrees	DIYmail	1	\$28.99	\$28.99	Yes	Amazon Link	4/4/2025	4/9/2025
10pcs High Voltage Generator DC 6-12V to 1000KV 1000000V	XIITIA	1	\$24.79	\$24.79	Yes	Amazon Link	4/4/2025	4/9/2025
Epoxy Resin Kit 72OZ	Puduo	1	\$36.99	\$36.99	Yes	Amazon Link	4/4/2025	4/9/2025
99.9% Solid Bare Copper Wire 18 Gauge	YEZHET	1	\$20.45	\$20.45	No	Amazon Link	4/4/2025	4/22/2025
SN74HC595N IC 8-Bit Shift Registers	Texas Instruments	1	\$7.35	\$7.35	Yes	Amazon Link	4/4/2025	4/9/2025
ELEGOO MEGA R3 Board (Arduino Mega Board)	ELEGOO	1	\$21.84	\$21.84	Yes	Amazon Link	4/4/2025	4/9/2025
5PCS GY-521 6 DOF MPU6050 Accelerometer Sensor Module	Stemedu	1	\$13.70	\$13.70	Yes	Amazon Link	4/4/2025	4/9/2025
M5 x 0.8 mm Thread, 12 mm Long Bolt (50 Count)	McMaster-Carr	1	\$5.05	\$5.05	No	McMaster-Carr Link	4/15/2025	4/21/2025
Class 04, M5 x 0.8 mm Thread Medium-Strength Steel Thin-Profile Hex Nut (100 Count)	McMaster-Carr	1	\$3.78	\$3.78	No	McMaster-Carr Link	4/15/2025	4/21/2025
M18 x 2.5 mm Thread, 45 mm Long Alloy Steel Socket Head Screw Black-Oxide (5 Count)	stemedu	6	\$12.76	\$78.56	No	McMaster-Carr Link	4/15/2025	4/21/2025
PETG 3D Printer Filament 1.75mm, (2.2lbs), (Orange)	OVERTURE 3D	2	\$21.99	\$43.98	No	Amazon Link	4/25/2025	
PETG 3D Printer Filament 1.75mm, (4.4lbs), (Black (2-Pack))	OVERTURE 3D	2	\$29.99	\$59.98	No	Amazon Link	4/25/2025	
			Total Spent:	\$475.02				
			Total:	\$475.02				
			Remaining Budget:	\$24.98				

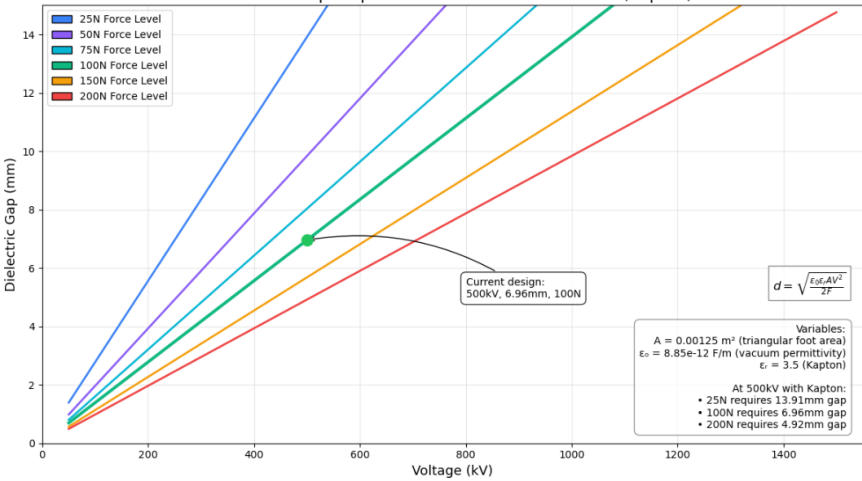
Force vs. Voltage with Kapton Dielectric (6.96mm Gap)



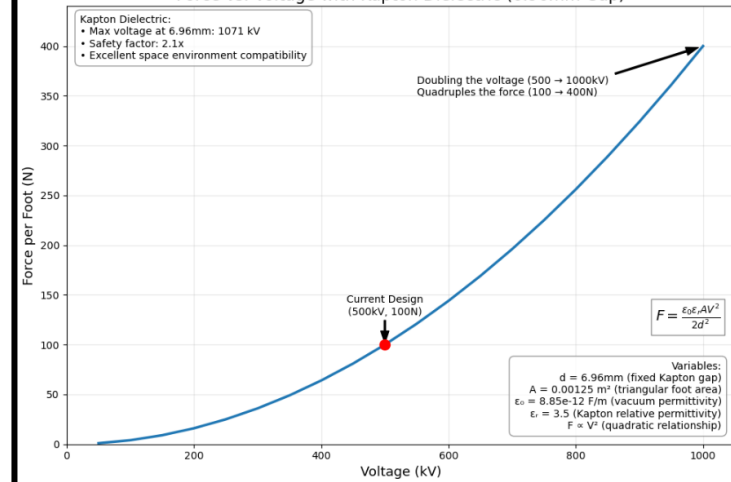
Vacuum vs. Kapton Dielectric Comparison (100N Force)



Dielectric Gap Required for Different Force Levels (Kapton)



Force vs. Voltage with Kapton Dielectric (6.96mm Gap)



Final Poster

COLLEGE OF ENGINEERING



Background

- **Unprecedented Core Exploration:** NASA's Psyche mission targets a unique metal-rich asteroid believed to be an exposed planetary core, offering scientists a never-before-possible look at what lies beneath planetary crusts.
- **Potential Metallic Treasure:** Asteroid 16 Psyche may consist of up to 90% metal (primarily iron and nickel), representing potentially one of the most valuable celestial bodies in our solar system and a window into planetary formation.
- **Extreme Environment Challenges:** Engineers must overcome exceptional difficulties including minimal gravity (0.144 m/s²), dramatic temperature variations (1-200K), and the unprecedented metallic surface requiring innovative attachment solutions.

Fixture Design

- High-temp resistant materials, Titanium/Aluminum 6061 for structural components
- Specialized dual-action anchoring (electrostatic + thermite bonding)
- Designed for extreme thermal cycling (1-200K)
- System designed to handle height differences up to 3 feet and 34 degrees of tilt

Operational Phases

- **Transit Configuration:** Legs folded for compact storage during journey
- **Deployment Configuration:** Legs extend and prepare for surface contact
- **Anchored Configuration:** Two-stage process:
 - Initial: Electrostatic adhesion provides temporary secure hold
 - Permanent: Thermite welding creates final bond



Oregon State University

Mechanical, Industrial, and Manufacturing Engineering

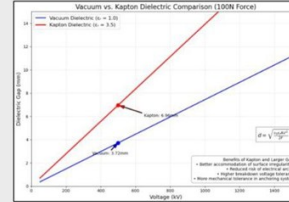
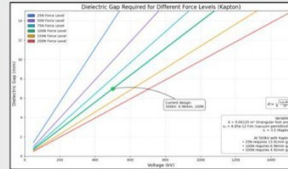
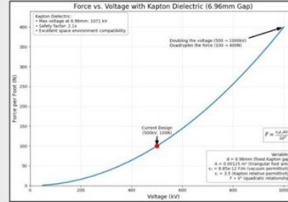
MIME5504A

NASA Psyche Space Elevator Anchor

Project Mission

Psyche is an asteroid with vast valuable resources, and we aim to develop an **anchor** for a space elevator tether to facilitate **sample transfer** between the asteroid and a geosynchronous orbiting spacecraft. Our team's primary goal is to create a well-designed anchoring system that can effectively secure to Psyche's unique ferrous surface in **microgravity conditions**. The system must adapt to **irregular terrain**, withstand **extreme temperature variations**, and provide sufficient **anchoring force** to support space elevator operations.

Electrostatic Anchor Design



- Anchor design based on Coulomb's law and the electrostatic force field generated between **parallel plate capacitors**.
- The graphs show the relationship between the **gap required** to produce an electric force strong enough to anchor to metallic surface of Psyche and the **input voltage**.
- The total anticipated anchoring **force per leg is 900 N = 202 lbf** for an overall system anchoring force of **808 lbf**.

System Features

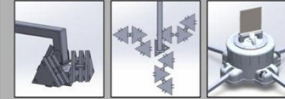
- **Central Mechanized Hub:** Houses motor and gear system with innovative clutch mechanism that independently controls each leg's position
- **Leg Modules:** Feature ball hinges and clutch mechanisms for flexibility and motion range to navigate irregular surface topography
- **Adaptable Feet Assembly:** Interconnected triangular segments that conform to Psyche's uneven metallic terrain, maximizing contact area
- **Tether Attachment Gimbal:** Flexible interface between anchoring system and space elevator ribbon

Primary References

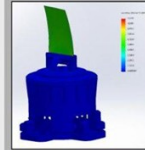
- ASTM International. (2015). ASTM E595-15, Standard Test Method for Total Mass Loss and Collected Volatile Condensable Materials from Outgassing in a Vacuum Environment.
- NASA. (2014). NASA-STD-0001, Structural Design and Test Factors of Safety for Spaceflight Hardware.
- NASA. (2016). NASA-STD-8016, Standard Materials and Processes Requirements for Spacecraft.
- NASA. (2014). NASA-STD-5017, Design and Development Requirements for Mechanisms.
- Viole, A., Ceriotti, M., & McInnes, C. R. (2021). Dynamics of a Nonrigid Orbital Siphon at a Near-Earth Asteroid. *Journal of Guidance, Control, and Dynamics*, 44(1), 137-148



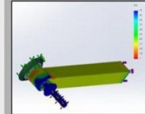
Preliminary Design



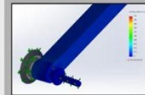
Finite Element Analysis



Central Mechanized Hub and Tether Attachment Gimbal stress distribution under 10 tons of upward tether force. The simulation visualizes von Mises stress concentrations (measured in MPa) across the device's structure with a FoS greater than 4. Tether simulated using titanium to approximate tensile strength similar to carbon composite material anticipated for the actual tether.



Fatigue Factor of Safety of Leg Axle Assembly showing 0.5 tons of Tensile Load Applied. Areas where FoS < 2 implies future reiterations of design.



von Mises stress concentrations of Leg Axle Assembly showing 0.5 tons of Tensile Load Applied. Leg stress does not exceed yield strength. Deformation is recoverable implying long cycle life.

Team Members

- **Project Manager:** Kyle Ladda-Levarien
- **CAD/Design Lead:** Jarel Shahim
- **Manufacturing Lead:** Kiefer Gonsalves
- **Robotics Lead:** Zachary Thomas
- **Faculty Advisor:** Sarah Oman
- **Sponsor:** NASA Psyche Mission

This work was created in partial fulfillment of Oregon State University Capstone Course MIME 488. The work is a result of the Psyche Student Collaborations component of NASA's Psyche Mission (<https://science.nasa.gov/missions/psyche/>). Psyche is a discovery mission to explore the asteroid 16 Psyche. The Psyche Mission is part of the NASA Discovery Program mission to solar system targets. Trade names and trademarks of ASU and NASA are used in the work for identification only. The images does not constitute an official endorsement or either expressed or implied. Oregon State University or its affiliates does not necessarily represent the official views of ASU or NASA.

How we Addressed Engineering Challenges

- Electrostatic Gripper:
 - Continued research on high voltage systems and electroadhesion
 - How to simulate outside of trial and error
- Robotics & Controls:
 - Wiring and test
 - Integrating/Programming servo motors & microcontrollers
- 3D Print Quality:
 - Standardization of parameters & outlined order of operations
 - Making sure tasks are equally distributed
- Structure Failure Points:
 - Simulated FEA of parts interfaces under a cyclic temperature range and static loading schemes.
 - Explore part life expectancy through fatigue analysis

How we met Engineering Specifications & Customer Requirements

Conclusion

- **Project Successes & Learning**

- **Key Takeaways:**



- **Next Capstone Project:**

- Iterative design improvements (Gear Design, Feet Design, Motors, Control System/Wiring)
 - Move towards advanced 3D prints after next iteration
 - Research of material selection, stress/strain calculations, Thermite Welding & Electromagnetic Adhesion proofs & calculations, controls, and CFD

Thank you! Questions?